WebClient Computing

CSS Basics

Department of Computer Science

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Yesterday, my inbox…

DaEunKim	**[kmu-cs-web-client-161] Homework 1 (#47)** - You can view, comment on, or merge this	View Pull Request	6:24 pm
iamkyungyong (2)	**[kmu-cs-web-client-161] fork test (#19)** - You can view, comment on, or merge this pull	View Pull Request	6:23 pm
EomHyeongGeun	**[kmu-cs-web-client-161] fork test2 (#46)** - fork fork You can view, comment on, or merge	View Pull Request	6:07 pm
SeungWooJeong	**[kmu-cs-web-client-161] the homework1 (#45)** - Hi! You can view, comment on, or merg	View Pull Request	5:55 pm
qr96	**[kmu-cs-web-client-161] Master (#44)** - hehe You can view, comment on, or merge this	View Pull Request	5:54 pm
Brilliantw	**[kmu-cs-web-client-161] Homework 1(20153173) (#43)** - You can view, comment on, or	View Pull Request	5:52 pm
knagwonji (3)	**[kmu-cs-web-client-161] homework (#41)** - You can view, comment on, or merge this pu	View Pull Request	5:51 pm
leeseokin (2)	**[kmu-cs-web-client-161] Merge remote-tracking branch 'refs/remotes/origin/master'**	View Pull Request	5:50 pm
sshun21	**[kmu-cs-web-client-161] Homework 1 (#42)** - You can view, comment on, or merge this pull request online at: https		5:50 pm
imgyeonghun	**[kmu-cs-web-client-161] 20121887 test (#40)** - You can view, comment on, or merge this	View Pull Request	5:48 pm
gwangjin	**[kmu-cs-web-client-161] first summary (#39)** - summary You can view, comment on, or	View Pull Request	5:47 pm
dokyeong2	**[kmu-cs-web-client-161] 0324 (#38)** - edit1 You can view, comment on, or merge this pul	View Pull Request	5:47 pm
KaileyR	**[kmu-cs-web-client-161] 20123196 Master (#37)** - You can view, comment on, or merge	View Pull Request	5:47 pm
kwc2013	**[kmu-cs-web-client-161] myfirsttest (#34)** - thank you You can view, comment on, or merge this pull request online		5:47 pm
ksm357	**[kmu-cs-web-client-161] 20132368 Fork (#33)** - homework! You can view, comment on, or merge this pull request c		5:47 pm

About Homework and GitHub Usage in the Course

- Some GitHub features remain
  - Pull from a parent repository, understanding conflicts
  - I will not cover them in the course
- GitHub tutorial sessions after the Thursday course (from 5:45)
  - From the next week around 30 minutes whenever needed
- Homework, project submission is OK for
  - GitHub (highly recommended)
  - cyber2010.kookmin.ac.kr
  - Note class attendance/participation takes 5%
- The real first homework about HTML will be assigned at the next Thursday

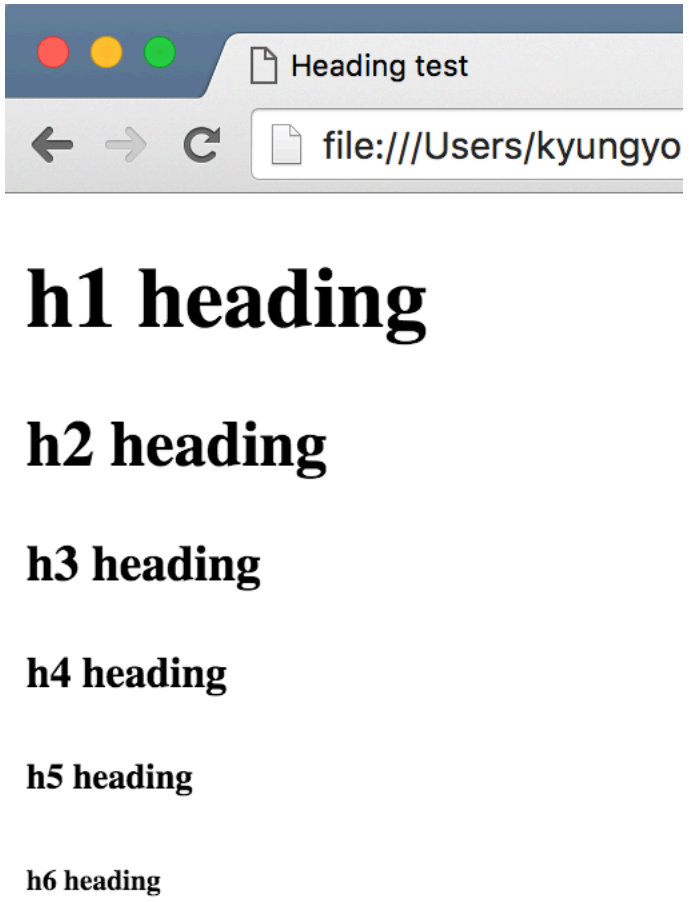
CSS – Cascading Style Sheets

- Decides how HTML file is presented in a web browser
- Remember <h1> ~ <h6> example
- <h1> ~ <h6> means hierarchical content organization
- It does not specify font size and color!
- Using CSS we can change the format

HTML Headings

- Organizing contents hierarchically
  - Recommended to use in order
  - h1, h2, h3, h4, h5, h6
  - Space is automatically added after heading
  - Search engine use headings to **index** the structure of the content

```html
<!DOCTYPE html>
<html>
<head>
  <title>Heading test</title>
</head>
<body>
  <h1>h1 heading</h1>
  <h2>h2 heading</h2>
  <h3>h3 heading</h3>
  <h4>h4 heading</h4>
  <h5>h5 heading</h5>
  <h6>h6 heading</h6>
</body>
</html>
```



Same HTML, Different Outlook

h1 heading

h2 heading

h3 heading

h4 heading

h5 heading

h6 heading

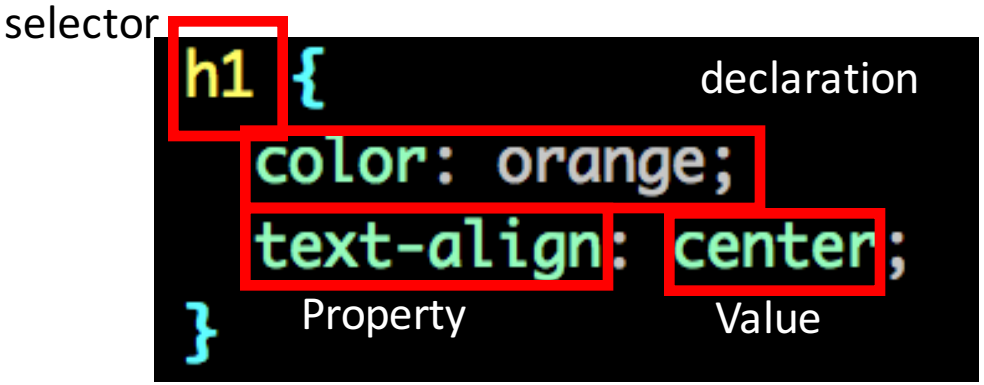
```html
<!DOCTYPE html>
<html>
<head>
  <style>
    h1 {
      color: orange;
      text-align: center;
    }
    h2 {
      color: blue;
      text-align: left;
    }
    h3 {
      color: green;
      text-align: left;
      font-size: xx-large;
    }
  </style>
  <title>Heading test</title>
</head>
<body>
  <h1>h1 heading</h1>
  <h2>h2 heading</h2>
  <h3>h3 heading</h3>
  <h4>h4 heading</h4>
  <h5>h5 heading</h5>
  <h6>h6 heading</h6>
</body>
</html>
```

Benefits of CSS

- Separation of content from its presentation – modular
- Reusable style by having an external CSS file
- Useful to show HTML in different design and layout
  - Desktop version and mobile device version

CSS Syntax

- CSS Syntax consists of
  - Selector – HTML elements where the style will be applied to
    - Tag, ID (starts from #), class (starts from .)
  - Declaration – specifies style to be applied to the selector. Multiple attributes separated by semi-colon;
    - Property: The name of the attribute (ex. font-size)
    - Value: The value of the attribute (ex. xx-large)
    - Property and Value is separated by colon



How to Insert Style Sheets

- External
  - An external style sheet file is created with an extension of .css and is referenced within head

```
<!DOCTYPE html>
<html>
<head>
 <link rel="stylesheet" type="text/css" href="styles.css">
  <title>My first web page</title>
</head>
```

  - <link> : a tag to define relationship (link) of two documents
    - An empty element – no value and attributes only
    - rel attribute – defines relation of linked documents
      - stylesheet – the linked document is a style sheet file
      - Other rel values - http://www.w3schools.com/tags/att_link_rel.asp

External Style Sheet Example

- Create a styles.css file and write some style values

```
h1 {
  color: orange;
  text-align: center;
}
```

- Link the styles.css file in a HTML file

```
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" type="text/css" href="styles.css">
  <title>My first web page</title>
</head>
<body>
  <h1>H1 heading</h1>
</body>
</html>
```

Name	Date Modified	Size	Kind
css-external.html	9:25 PM	182 bytes	HTML
styles.css	9:25 PM	46 bytes	CSS style sheet

How to Insert Style Sheets

- Internal
  - Inserted within header <style> tag

```html
<!DOCTYPE html>
<html>
<head>
  <style>
    h1 {
      color: orange;
      text-align: center;
    }
  </style>
  <title>Heading test</title>
</head>
```

- Inline styles
  - A style can be added for a single element within a start tag

```html
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
</head>
<body>
  <h1 style="color: orange;text-align: center;">h1 heading</h1>
</body>
</html>
```

Using External CSS File is an Ideal Way

- Benefits of external CSS
  - Share a CSS file across different HTML pages (reusable)
  - Reduce HTML file size
    - This can be especially beneficial when the content (HTML) update happens frequent while a CSS update happens rarely
    - A browser can cache CSS file to reduce network overhead
    - Lower loading time (note CSS file can be cached)

Characters of Using Internal CSS File

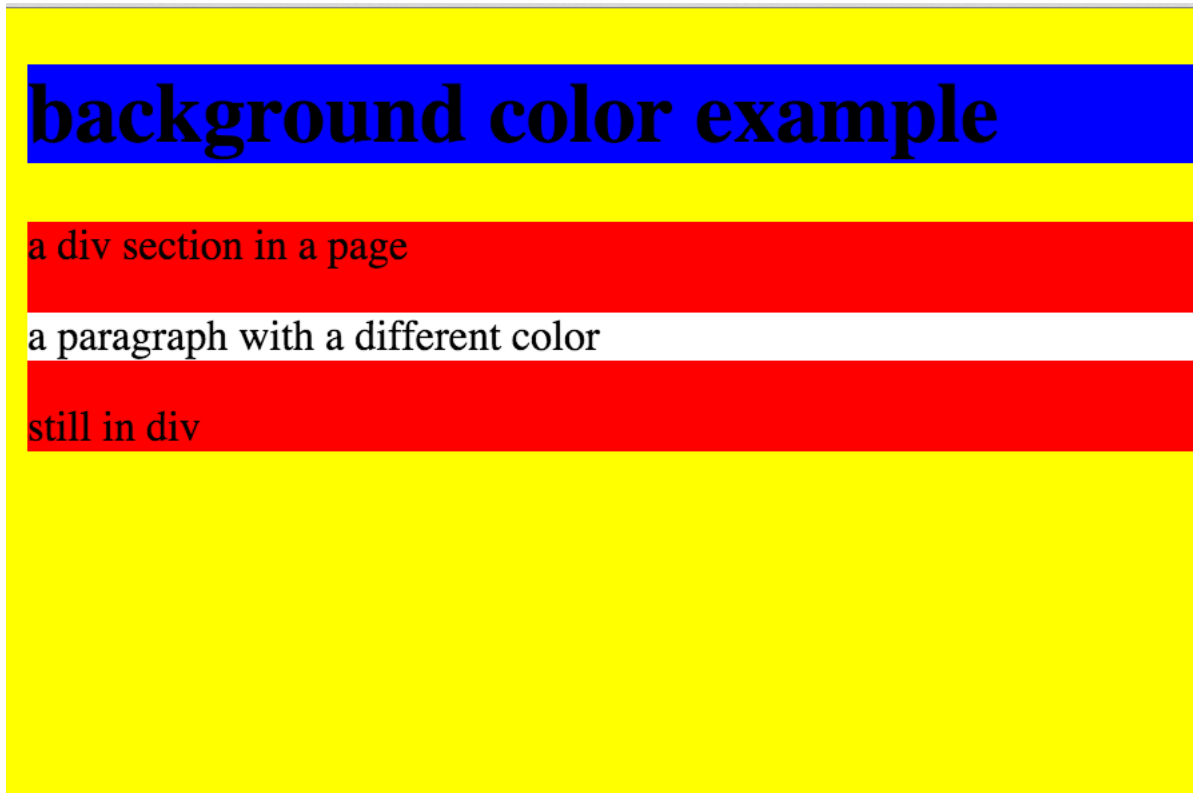
- Benefits of internal CSS
  - One style of same element – note selector can be tag name, class
  - Might be easier to work as style and HTML is in a sinlge file
- Disadvantage of internal CSS
  - If same style applies to multiple HTML documents, the style has to be copied
  - File size can be large

Characters of Using Inline CSS

- Note that the inline CSS has the highest priority
- Benefits of inline CSS
  - Good for test – can easily try a new feature
- Disadvantages of inline CSS
  - Overriding internal and external CSS file – a page can render in an unexpected way
  - Has to be applied for every element

Setting Background

- background-color : sets background color of a page

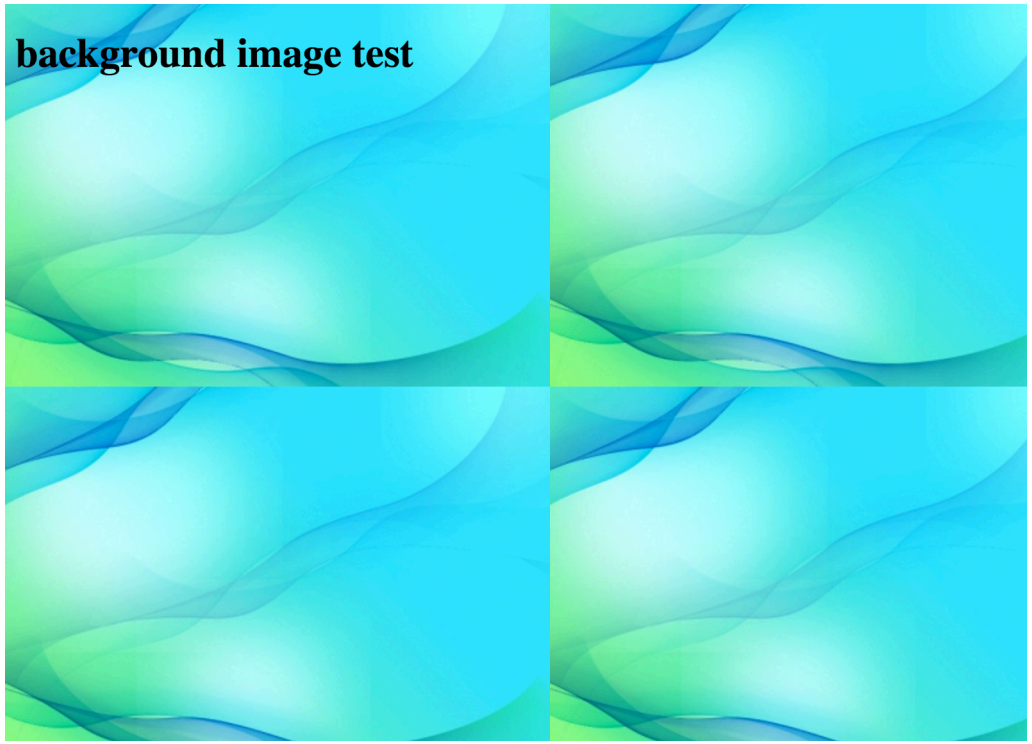


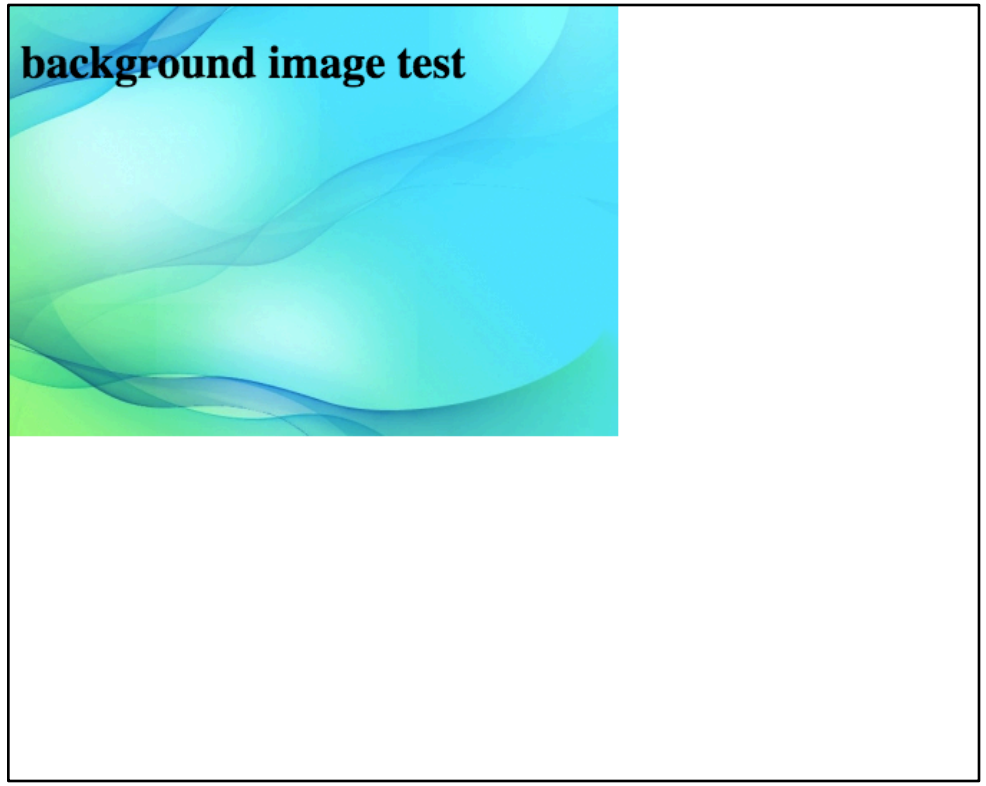

```html
<!DOCTYPE html>
<html>
<head>
<style>
  body {
    background-color: yellow;
  }
  h1 {
    background-color: blue;
  }
  div {
    background-color: red;
  }
  p {
    background-color: white;
  }
</style>
</head>
<body>

<h1>background color example</h1>
<div>
a div section in a page
<p>a paragraph with a different color</p>
still in div
</div>

</body>
</html>
```

Setting image Background

- background-image
  - Use an image to use as a background
  - By default, an image is repeated to cover the entire area

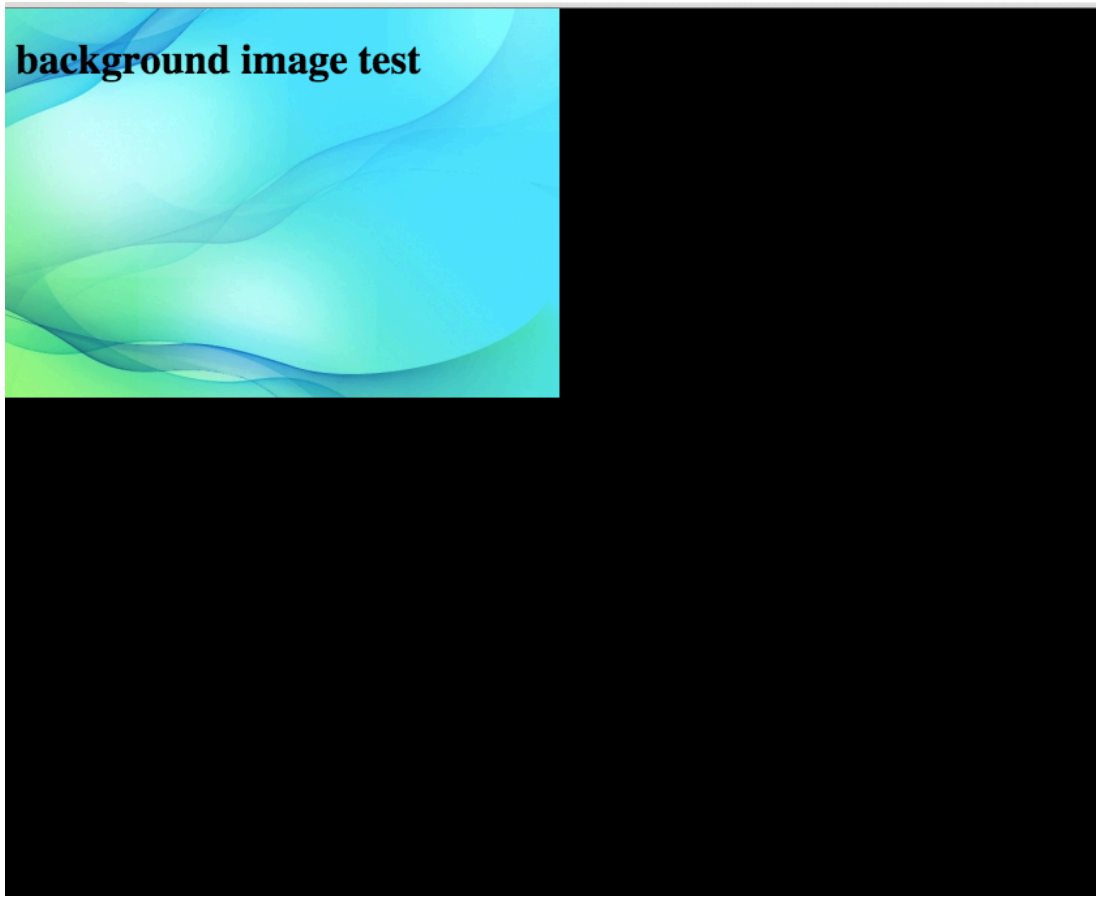
```html
<!DOCTYPE html>
<html>
<head>
<style>
  body {
    background-image: url("bg_image.jpg");
    background-repeat: no-repeat;
  }
</style>
</head>
<body>
<h1> background image test </h1>
</body>
</html>
```





Combination of Color and Image

- Combination of color and image is possible




```html
<!DOCTYPE html>
<html>
<head>
<style>
  body {
    background-image: url("bg_image.jpg");
    background-repeat: no-repeat;
    background-color: black;
  }
</style>
</head>
<body>
<h1> background image test </h1>
</body>
</html>
```

Class Selector in CSS

- Class in CSS
  - An attribute to group a set of tags with a similar effect
  - In CSS, it starts with dot (.) followed by a class name

```html
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    .red_background {
      background-color: red;
    }
  </style>
</head>
<body>
  <p class="red_background">CSS using class</p>
</body>
</html>
```

CSS using class

CSS Borders

- Specifies a style of a border of a given element

A dotted border.

A dashed border.

A solid border.

A double border.

A groove border.

A ridge border.

An inset border.

An outset border.

No border.

```html
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    .dotted {border-style: dotted;}
    .dashed {border-style: dashed;}
    .solid {border-style: solid;}
    .double {border-style: double;}
    .groove {border-style: groove;}
    .ridge {border-style: ridge;}
    .inset {border-style: inset;}
    .outset {border-style: outset;}
    .none {border-style: none;}
  </style>
</head>
<body>
  <p class="dotted">A dotted border.</p>
  <p class="dashed">A dashed border.</p>
  <p class="solid">A solid border.</p>
  <p class="double">A double border.</p>
  <p class="groove">A groove border.</p>
  <p class="ridge">A ridge border.</p>
  <p class="inset">An inset border.</p>
  <p class="outset">An outset border.</p>
  <p class="none">No border.</p>
</body>
</html>
```

Border Color and Width

- Border's color and width can be set using
  - border-color
  - border-width

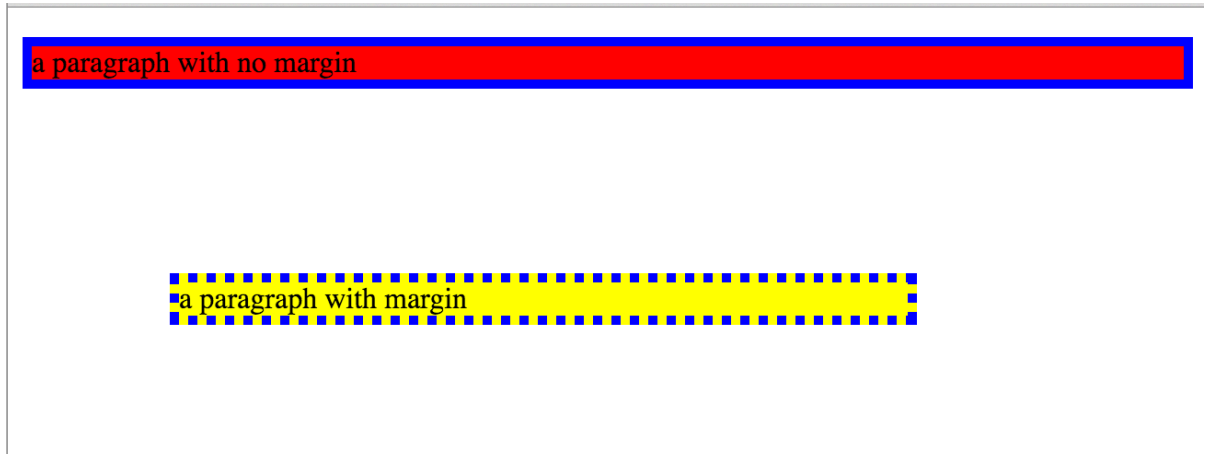
---

border color width test

```
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    h3 {
      border-style: solid;
      border-color: red;
      border-width: 10px;
    }
  </style>
</head>
<body>
  <h3> border color width test </h3>
</body>
</html>
```

CSS Margins

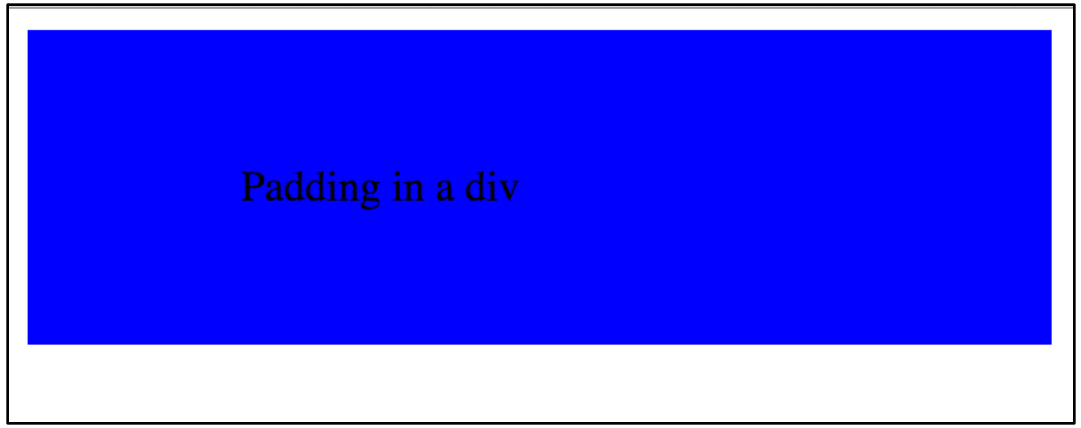
- Margins allows to specify the space outside of the given element




```html
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    .no_margin {
      background-color:red;
      border-width:5px;
      border-color: blue;
      border-style: solid;
    }
    .with_margin {
      background-color:yellow;
      border-width:5px;
      border-color: blue;
      border-style: dotted;
      margin-top: 100px;
      margin-bottom: 100px;
      margin-right: 150px;
      margin-left: 80px;
    }
  </style>
</head>
<body>
  <p class="no_margin">a paragraph with no margin</p>
  <p class="with_margin">a paragraph with margin</p>
</body>
</html>
```

CSS Padding

- Padding allows to specify the space between the border and the content in an element




```html
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    .padding {
      background-color: blue;
      padding-top: 50px;
      padding-right: 30px;
      padding-bottom: 50px;
      padding-left: 80px;
    }
  </style>
</head>
<body>
  <div class="padding">Padding in a div</div>
</body>
</html>
```

References

- http://www.w3schools.com/css/default.asp - CSS Tutorial